Incident COVID-19 infections before Omicron in the U.S.

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Abstract

The timing and magnitude of COVID-19 infections are of interest to the public and to public health, but these are challenging to ascertain due to the volume of undetected asymptomatic cases and reporting delays. Accurate estimates of COVID-19 infections based on finalized data can improve understanding of the pandemic and provide more meaningful quantification of disease patterns and burden. Therefore, we retrospectively estimated daily incident infections for each U.S. state prior to Omicron. To this end, reported COVID-19 cases were deconvolved to their date of infection onset using delay distributions estimated from the CDC line list. Then, a novel serology-driven model was used to scale these deconvolved cases to account for the unreported infections. The resulting infections incorporated variant-specific incubation periods, reinfections, and waning antigenic immunity. They clearly demonstrate that the reported cases failed to reflect the full extent of disease burden in all states. Most notably, infections were severely underreported during the Delta wave, with an estimated reporting rate as low as 6.3% in New Jersey, 7.3% in Maryland, and 8.4% in Nevada. Moreover, in 44 states, fewer than 1/3 of infections appeared as case reports, and there were sustained periods where surges in infections were virtually undetectable through reported cases. This pattern was clearly illustrated by North and South Dakota during the spring of 2021, as well as by several Northeastern states during the Delta wave of late summer that year. Therefore, while reported cases offered a convenient proxy of disease burden, they failed to capture the full extent of infections, and severely underestimated the true disease burden. This retrospective analysis also estimated other important quantities for every state, including variant-specific deconvolved cases, time-varying case ascertainment ratios, as well as infection-hospitalization ratios and infection-fatality ratios.
Keywords: COVID-19; SARS-CoV-2; Infections; Deconvolution; Time series; Seroprevalence; Antibody

1 Introduction

Reported COVID-19 cases were a staple in tracking the pandemic at varying geographic resolutions (Dong et al., 2020; The New York Times, 2020; The Washington Post, 2020). Yet, for every case that was eventually reported to public health, several infections were likely to have occurred. To see why, it is important to understand *whose* cases were being reported and what differentiates them from unreported cases as well as *when* these case reports happen. Figure 1 shows an idealized path of a symptomatic infection that is eventually reported to public health. This figure illustrates a number of sources of bias in the reporting pipeline. For instance, diagnostic testing mainly targeted symptomatic individuals; thus, infected individuals exhibiting little to no symptoms were omitted (Centers for Disease Control and Prevention, 2022). In addition, testing practices, availability, and uptake vary temporally and spatially (European Centre for Disease Prevention and Control, 2020; Hitchings et al., 2021; Pitzer et al., 2021). Finally, cases provided a belated view of the pandemic's progression, because they were subject to delays due to the viral incubation period, the speed and severity of symptom onset, laboratory confirmation, test turnaround times, and eventual submission to public health (Pellis et al., 2021; Washington State Department of Health, 2020). For these reasons, reported cases were lagging indicators of the course of the pandemic. Furthermore, they did not represent the actual number of new infections that occur on any given day based on exposure to the pathogen. Since there was no large-scale surveillance effort in the United States that reliably tracked symptom onset, let alone infection onset, ascertaining the onset of all *infections* is challenging.

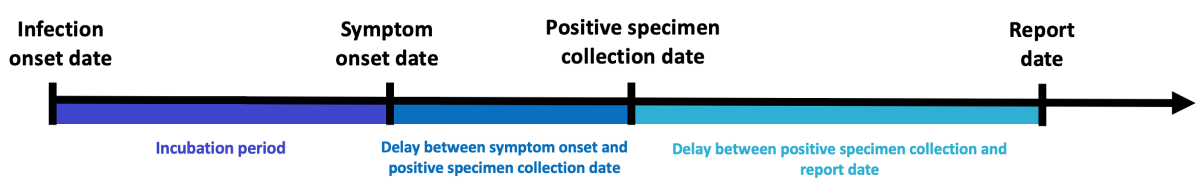

Figure 1: Idealized chain of events from infection onset to case report date for a symptomatic infection that is eventually reported to public health.

Infection prevalence studies were conducted in various countries over the course of the pandemic to measure and monitor the prevalence of COVID-19 in real- time. The UK was a leader in such large-scale

COVID-19 surveillance studies with the REACT-1 programme, which tracked the pandemic through at-home testing of randomly selected individuals across England (Imperial College London, 2022). In addition, the ONS COVID-19 Infection Survey, took the at-home testing approach to households across the UK to ascertain how many people presently and previously had COVID-19 (Office for National Statistics, 2020). Comparable studies were conducted in other countries, including France (Vaux et al., 2023), Germany (Jacob et al., 2021), Hungary (Merkely et al., 2020), and Iceland (Gudbjartsson et al., 2020), though many of these efforts were smaller-scale and shorter-term. The U.S. supported some smaller-scale studies like the Community Prevalence of SARS-CoV-2 Study (COMPASS), which assessed the population-based prevalence of COVID-19 infection in over 22,000 individuals from 15 select communities across the U.S. in 2021 (Justman et al., 2024), as well as the Digital Engagement and Tracking for Early Control and Treatment (DETECT) study, which collected over 35,000 self-reported test results from 2020 to 2022 (Kolb et al., 2023). While these studies provided valuable insights, their voluntary nature and the resulting participation bias meant they more likely captured emerging cases, and therefore, tended towards incidence rather than true prevalence. Furthermore, large-scale studies like REACT-1 or ONS are costly due to extensive sampling and mass testing (Pavelka et al., 2021). Alternatives such as seroprevalence sampling may be more sustainable over time due to its focus on routine blood sampling, and less resource-intensive than traditional PCR-based testing (COVID-19 Immunity Task Force, 2023). While infection prevalence studies are important for real-time monitoring, their cost, cross-sectional nature, and logistical complexity limit their feasibility for long-term tracking and retrospective analysis. Consequently, leveraging routine data streams already integrated with existing infrastructure can give a more clear and consistent view of the pandemic's progression, while enhancing transparency and public participation.

Contextualizing the course of the pandemic, understanding the effects of interventions, and drawing insights for future pandemics is challenging because the spatial and temporal behaviour of infections is unknown. While reported cases provide a convenient proxy of the disease burden in a population, it is still incomplete, delayed, and misrepresents the true size and timing of the pandemic. Regardless of these difficulties, it is important to the public and to public health to perform a pandemic post-mortem. Estimates of daily incident infections are one such way to measure this and can guide understanding of the pandemic burden over space and time.

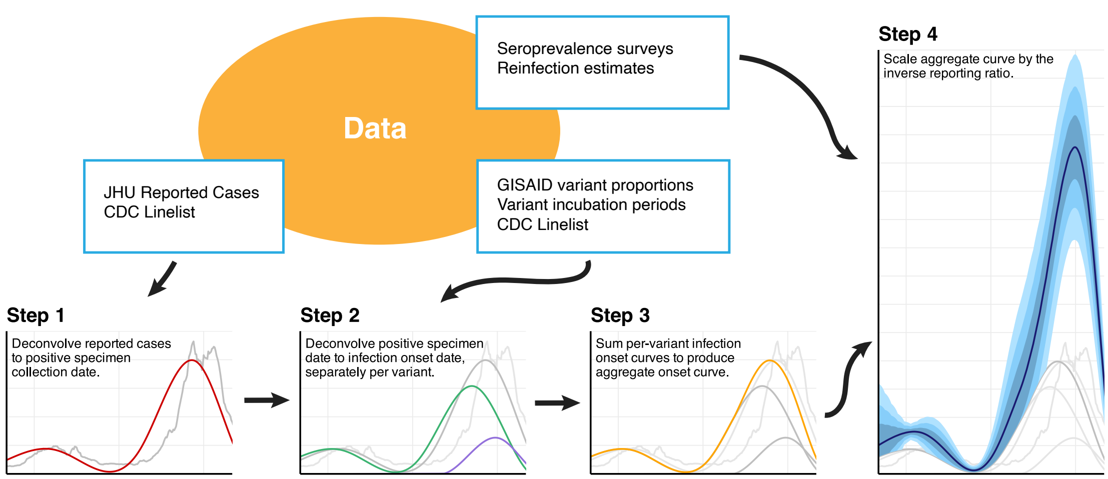

Figure 2: Flowchart of the data and major analysis steps required to get from reported cases to incident infection estimates. In Step 1, we used the CDC line list data to deconvolve (i.e. push back using estimated probabilities) reported cases (grey) to the date of positive specimen (red). Step 2 separately deconvolved these to the date of infection by variant (Epsilon in Purple, Ancestral in Green), before summing across all variants (orange) in Step 3. Finally, we used seroprevalence survey and time-varying reinfection data to account for the unreported infections. In the final infection estimates, note that the uncertainty comes solely from the seroprevalence step, while the other steps contribute deterministically.

In this work, we provide a data-driven reconstruction of daily incident infections for each U.S. state before the onset of Omicron. Using state-level line list data, we estimated state-date specific distributions for the delay from symptom onset to positive specimen date and positive specimen to case report date. We combined these with variant-specific incubation period distributions to deconvolve daily reported COVID-19 cases back to their infection onset, removing the effects of the delays. Finally, we adjusted for unreported infections with seroprevalence and reinfection data, accounting for the waning of antigenic immunity over time. A graphical depiction of our procedure is shown in Figure 2. We also provide a brief description of the key data sources used in our procedure in Appendix A.

Our results focus on the features of the infection estimates and the implications of using them, rather than reported cases, to assess the impact of the pandemic in U.S. states. We also calculated simple time-varying infection-hospitalization ratios (IHRs) and infection-fatality ratios (IFRs) for each state and compared them with their case-based counterparts: the case-hospitalization ratios (CHRs) and case-fatality ratios (CFRs). While these analyses provide a glimpse into the utility of our infection estimates, we believe that there is much more to be explored, and we hope that our work serves as a benchmark for future retrospective analyses.

2 Methods

In what follows, we describe how we estimated the daily incident infections for each U.S. state from June 1, 2020 to November 29, 2021. Figure 2 summarizes the major analysis tasks. First, we estimated the delays from positive specimen to report date and used them to push back the reported cases to their sample collection dates. Next, we estimated the delay from symptom onset to sample collection, combined this with variant-specific infection-to-symptom delays, and used these to push back the cases to infection onset. The resulting case estimates were aggregated across variant categories and adjusted by the case ascertainment ratio, estimated with seroprevalence survey data and a model for antigenic immunity.

2.1 From reported cases to positive specimen collection

Deconvolution “pushes back” reported cases to the likely date of positive specimen collection. An important aspect of our methods is that deconvolution is not the same as a simple shift, rather it involves the distribution of delays (specific to each state and date). Simply shifting cases back in time would fail to reflect the fact that some cases take much longer to be reported than others (Appendix B).

We will start by describing how the model for deconvolution inferred the likely dates of positive specimen collection from reported cases before describing how the CDC line list (Centers for Disease Control and Prevention, 2020a) was used to estimate the necessary delay distributions. Together, these are the ingredients for Step 1 in Figure 2. Define $y_{\ell,t}$ to be the number of new cases reported in location ℓ at time t, as reported by the John Hopkins Center for Systems Science and Engineering (JHU CSSE, Dong et al., 2020) and retrieved with the COVIDcast API (Reinhart et al., 2021). Let $\pi_{\ell,t}(k)$ be the probability that cases with positive specimen collection at time $t-k$ are reported at t. Then, we modelled $y_{\ell,t}$ as a Gaussian with mean

$$\mathbb{E}[y_{\ell,t} \mid x_{\ell,s}, s \leq t] = \sum_k \pi_{\ell,t-k}(k) x_{\ell,t-k}, \tag{1}$$

which is a probability weighted sum of the number of positive specimens collected k days earlier, $x_{\ell,t-k}$. We estimated $\mathbf{x}_\ell = \{x_{\ell,1}, \ldots, x_{\ell,T}\}$ by minimizing the negative log-likelihood with a penalty that encouraged smoothness in time. Thus, our estimator was given by

$$\widehat{\mathbf{x}}_\ell = \operatorname*{argmin}_{\mathbf{x}} \sum_t \left(y_{\ell,t} - \sum_k \pi_{\ell,t-k}(k) x_{t-k} \right)^2 + \lambda \sum_t \left| x_t - 4x_{t-1} + 6x_{t-2} - 4x_{t-3} + x_{t-4} \right|. \tag{2}$$

The solution to this minimization problem was an adaptive piecewise cubic polynomial (Tibshirani, 2014, 2022) and could be accurately computed with ease (Jahja et al., 2022; Ramdas and Tibshirani, 2016). We selected the tuning parameter λ with cross-validation to minimize the out-of-sample reconvolution error.

To estimate the $\pi_{\ell,t}(k)$ for all states ℓ, times t, and delays k, we used the CDC line list (Centers for Disease Control and Prevention, 2020a). The line list contained three key dates of interest for many cases that eventually appeared in case reports: symptom onset, positive specimen collection, and report to the

CDC. Handling missingness in these dates required careful attention (Appendix C). Define $z_{\ell,t}$ to be a case report occurring at time t in location ℓ. We assumed that positive samples were reported within 60 days and that no test was reported on the same date as it was collected. Under these assumptions, let $N_{\ell,t}$ be the total number of $z_{\ell,r}$ with positive specimen collection date r in a window $r \in [t-75+1, t+60]$ around t. Then, we computed the observed probability mass function (pmf)

$$\tilde{p}_{\ell,t}(k) = \frac{1}{N_{\ell,t}}\big(\#\ z_{\ell,r} \text{ with positive specimen at } r-k\big)\mathbf{1}(0 < k \leq 60), \tag{3}$$

where $\mathbf{1}(Z) = 1$ if Z is true and 0 otherwise. We also computed a similar national pmf, $\tilde{p}_t(k)\mathbf{1}(0 < k \leq 60)$, without restricting to location ℓ. Next, let $\alpha_{\ell,t}$ be the ratio of $N_{\ell,t}$ to the number of cases reported by JHU CSSE (Dong et al., 2020) in the window $[t-60+2, t+75]$. Then, we computed $p_{\ell,t} = \alpha_{\ell,t}\tilde{p}_{\ell,t} + (1-\alpha_{\ell,t})\tilde{p}_t$. This construction was adopted to allow for more reliance on the state estimate when a larger fraction of the JHU case reports appeared in the CDC line list. We calculated the mean $m_{\ell,t}$ and variance $v_{\ell,t}$ of the pmf $\{p_{\ell,t}(k)\}$ and estimated the best-fitting gamma distribution by solving the moment equations $m_{\ell,t} = \alpha_{\ell,t}\theta_{\ell,t}$ and $v_{\ell,t} = \alpha_{\ell,t}\theta_{\ell,t}^2$ for the shape $\alpha_{\ell,t}$ and scale $\theta_{\ell,t}$. Finally, we discretized the resulting gamma density to the original support to produce an estimate $\widehat{\pi}_{\ell,t}(k)$ of the delay distribution $\pi_{\ell,t}(k)$. Additional details are deferred to Appendix D.

2.2 From positive specimen collection to infection onset

To continue, pushing positive specimen collection time back to infection onset (Step 2 in Figure 2), we used a procedure very similar to that described above and specified in Equations (1) and (2). However, because the delays involve the time from infection to symptom onset, these were treated as variant-specific. We used our estimates from Section 2.1, $\widehat{\mathbf{x}}_\ell$, but we weighted them corresponding to the mix of variants in circulation. To estimate the daily proportions of the variants circulating in each state, we used GISAID genomic sequencing data from CoVariants.org (Elbe and Buckland-Merrett, 2017; Hodcroft, 2021), and estimated a multinomial logistic regression model. This procedure is standard (Appendix E) (Annavajhala et al., 2021; Figgins and Bedford, 2021; Obermeyer et al., 2022). The resulting estimated probability of variant j is given by $\hat{v}_{j\ell,t}$.

To estimate variant-specific delays from infection to positive specimen collection, we convolved the location-time-specific symptom-to-test distributions (that were estimated from the CDC line list in the same way as in Section 2.1), with variant-specific incubation periods. The convolution of these gave a distribution $\widehat{\tau}_{j\ell,t}(k)$. Details on the convolution and its inputs are in Appendices F.1–F.3.

Analogous to Equations (1) and (2), for each variant j, we modelled the variant-specific, deconvolved cases as Gaussian with mean

$$\mathbb{E}\left[\widehat{v}_{j\ell,t}\widehat{x}_{\ell,t} \mid u_{j\ell,s}, s \leq t\right] = \sum_k \widehat{\tau}_{j\ell,t-k}(k)u_{j\ell,t-k} \tag{4}$$

and estimated $\mathbf{u}_{j\ell}$ by minimizing the negative log-likelihood with a penalty to promote smoothness:

$$\widetilde{\mathbf{u}}_{j\ell} = \operatorname*{argmin}_{\mathbf{u}} \sum_t \left(\widehat{v}_{j\ell,t}\widehat{x}_{\ell,t} - \sum_k \widehat{\tau}_{j\ell,t-k}(k)u_{t-k}\right)^2 + \lambda \sum_t \big|u_t - 4u_{t-1} + 6u_{t-2} - 4u_{t-3} + u_{t-4}\big|. \tag{5}$$

Note that the ℓ_1 penalty, applied to small, adjacent subsets of deconvolved cases, results in piecewise polynomials with adaptively chosen segment locations. This enables the model to capture localized changes and promote smoothness within segments, in contrast to the ℓ_2 penalty used in smoothing splines that enforces global smoothness. The smoothing occurs over sets of five consecutive case counts (roughly weekly smoothing) as determined by the discrete derivative order.

We call the solution $\widetilde{\mathbf{u}}_{j\ell}$ the *variant-specific deconvolved cases* and emphasize that these are cases that were eventually reported to public health. Because this deconvolution was performed separately for each location and variant, we summed over the variants at each time t, and denote the total deconvolved cases at location ℓ as $\widehat{\mathbf{u}}_\ell = \sum_j \widetilde{\mathbf{u}}_{j\ell}$ (Step 3 in Figure 2). Note that these deconvolved cases were indexed by the time of infection onset rather than case report.

2.3 Inverse reporting ratio and the antibody prevalence model

To capture the unreported infections, it was necessary to adjust the deconvolved case estimates by the inverse reporting ratio, which was the ratio of the number of incident infections to incident reported infections (Step 4 in Figure 2). Seroprevalence of anti-nucleocapsid antibodies represented the percentage of people who had at least one resolving or past infection (Centers for Disease Control and Prevention, 2020b), so we developed a model that used the change in subsequent seroprevalence measurements to estimate all new infections. We used two seroprevalence surveys to estimate the proportion of the population with evidence of previous infection in each state over time Blood Donor Seroprevalence Survey and the Nationwide Commercial Lab Seroprevalence Survey (Appendix G) (Centers for Disease Control and Prevention, 2021a,b).

To account for different surveys occurring on different dates with roughly weekly availability and measurement error, we treated actual seroprevalence $s_{\ell,m}$ as a latent variable available on Monday (using m rather than t to denote Mondays). Therefore, the observed seroprevalence survey measurements r^1_m and r^2_m were modelled as Gaussian,

$$r^1_{\ell,m} \mid s_{\ell,m},\ w^1_{\ell,m} \sim \mathrm{N}(s_{\ell,m},\ w^1_{\ell,m}\sigma^2_{\ell,r}), \tag{6}$$

$$r^2_{\ell,m} \mid s_{\ell,m},\ w^2_{\ell,m} \sim \mathrm{N}(s_{\ell,m},\ w^2_{\ell,m}\sigma^2_{\ell,r}), \tag{7}$$

with source-specific measurement errors, $w^1_{\ell,m}$ and $w^2_{\ell,m}$, that were scaled proportional to the reported uncertainty.

To complete the model, we assumed that latent seroprevalence could be modelled as Gaussian with mean given by a fraction of the previous seroprevalence measurement at m plus the reinfection-adjusted deconvolved cases multiplied by the inverse reporting ratio at time m:

$$\mathbb{E}[s_{\ell,m+1} \mid s_{\ell,m}] = (1-\gamma)s_{\ell,m} + a_{\ell,m}(1-z_m) \sum_{t\in[m,m+1]} \widehat{u}_{\ell,t}, \tag{8}$$

where $\widehat{u}_{\ell,t}$ are deconvolved cases (from Section 2.2), z_m is the fraction of reinfections, and $a_{\ell,m}$ is the inverse reporting ratio. Note that γ is the fraction of people whose level of infection-induced antibodies fell below the detection threshold between time t and time $t+1$. The daily fraction of new infections, z_t, was based on surveillance work conducted by the Southern Nevada Health District (Ruff et al., 2022), and these estimates were broadly similar to those in other locations with available data (Hawaii Department of Health, 2022; New York State Department of Health, 2023; Ruff et al., 2022; Washington State Department of Health, 2022). Finally, we specified the time-varying evolution of the inverse reporting ratio as Gaussian with expectation,

$$\mathbb{E}[a_{\ell,m+1} \mid a_{\ell,m},\ a_{\ell,m-1},\ a_{\ell,m-2}] = 3a_{\ell,m} - 3a_{\ell,m-1} + a_{\ell,m-2}. \tag{9}$$

This construction for Equation (9) resulted in estimates that varied smoothly in time.

The antibody prevalence model specified by Equations (6) to (9) is a state space model with latent variables $\mathbf{s}_\ell$ and $\mathbf{a}_\ell$. In this way, the latent variables and all unknown parameters could be estimated using maximum likelihood, despite missing or irregularly-spaced survey measurements. Additionally, latent quantities could be extrapolated beyond the times of measured seroprevalence. Additional details of this methodology and the computation of the associated uncertainty measurements are in Appendix H.

2.4 Lagged correlation to hospitalizations and time-varying IHRs and IFRs

From the COVIDcast API (Reinhart et al., 2021), we retrieved the daily number of confirmed COVID-19 hospital admissions for each state that were collected by the U.S. Department of Health and Human Services (HHS). We used our infection estimates $\widehat{\mathbf{u}}_\ell$ to compute the lagged correlation with hospitalizations. The goal of this analysis was to find the lag between infection and hospitalization rates that gave the highest average rank-based correlation across U.S. states. Thus, we considered a wide range of possible lag values ranging from 1 to 25 days. Then, to assess the impact of our modelling choices, particularly the contribution of the main steps to the lagged correlation analysis, we conducted an ablation study that is detailed in Appendix I.

For each considered lag, we calculated Spearman's correlation between the daily state infection and hospitalization rates from June 1, 2020, to November 29, 2021, using a center- aligned rolling window of 61 days. We then averaged these correlations across all states and times for each lag.

The lag that led to the highest average correlation was then used to estimate the time-varying IHRs for each state. Each IHR can be computed by dividing the number of individuals who were hospitalized due to COVID-19 by the estimated total number of those who were infected on the lagged number of days before. However, to stabilize these lagged IHR estimates, we used the sum of hospitalizations and infections within a window of 61 days centered on the date of interest, rather than just using one pair of dates for each computation.

To evaluate the impact of window size on the resulting stabilized IHR and CHR estimates, we conducted a sensitivity analysis by varying the window size from 15 days to 91 days. The results of this analysis are reported in Appendix J.

The same procedure used for IHRs was utilized to compute IFRs, with a few data-driven adjustments. Firstly, we retrieved the number of weekly new deaths with COVID-19 for each state from COVIDcast API (Reinhart et al., 2021), collected by the National Center for Health Statistics (NCHS). We opted for these death counts over those from JHU CSSE because they are aligned by the date of death and not the report date. Since these are at the weekly level, while the infection estimates are daily, we converted these to daily by using a simple proportional allocation approach for distributing the weekly counts across the adjacent weeks. Firstly, we created daily estimates for each such weekly total by dividing by seven. For weeks that had a missing value, we used a simple forward fill approach to fill-in the missing value. Then, since each weekly total count of mortality data was situated on a Sunday, we estimated a value for each day in a week by using a weighted combination of the preceding Sunday's and the following Sunday's values. For instance, Monday's value was constructed from a weighted combination, using 6/7 of the preceding Sunday's value and 1/7 of the following Sunday's values. Similarly, for Tuesday, the weighted combination was comprised of 5/7 of the preceding Sunday's value and 2/7 of the following Sunday's values. This approach extended to other days of the week. The underlying idea was that the weight was adjusted based on the position of the day in the week, and weighting more towards the value of the Sunday that the day under consideration was closer to.

To identify the optimal lag, we considered a larger range of lag values than for hospitalizations, from 1 to 35 days, as deaths typically follow hospitalization. As with the IHRs, we performed an ablation study to evaluate the effect of our modelling choices (Appendix K).

Next, we computed Spearman's rank-based correlation between the state infection and death rates using a 91-day center-aligned rolling window. Then, we averaged across the states and times to produce a single correlation estimate for each lag. The lag that gave the highest correlation was used to compute time-varying IFRs. Similar to the approach for IHRs, we constructed time-varying IFRs by using a 91-day rolling window to sum the deaths and infections within that window. Finally, to account for the effect of window size on the resulting IFRs, we conducted a sensitivity analysis by varying the window size from 31 to 101 days, with the results provided in Appendix L.

3 Results

3.1 Infection estimates and cases-to-infections ratios across the U.S. states

Prior to Omicron, the largest infection outbreaks were observed in the late summer and early fall of 2021 in Louisiana, Georgia, Idaho, and Montana (Figures 3 to 4). During this time, the state with the highest rate of infections on a single day was Louisiana, with 476 infections per 100K (95% CI: 294, 658) on July 20, 2021. For comparison, the state's 7-day average case rate peaked at 126 cases per 100K on August 13, 2021. Idaho follows with an infections peak of 457 per 100K (95% CI: 319, 595) on September 7, 2021, and a case peak of 76 per 100K occurring shortly thereafter on September 13, 2021. The period of lowest viral transmission was in the summer of 2020, when Vermont had fewer than 10 infections per 100K per week from June to August, the longest such lull observed for any state.

Nearly all states exhibited two major waves in infections—the Ancestral wave began in the fall of 2020 and extended into the winter season, while the Delta wave started in the late summer of 2021 and continued into mid-fall. In general, greater similarities in the strength and magnitude of outbreaks emerged in small clusters of states that border each other (Idaho and Montana; North and South Carolina), which presented waves of infections that mirrored each other in amplitude and timing.

While the Ancestral, Alpha, and Delta waves were visible for most states, there were clear outbreaks in unreported infections that were not easily detectable from cases alone. For example, a wave of infections was

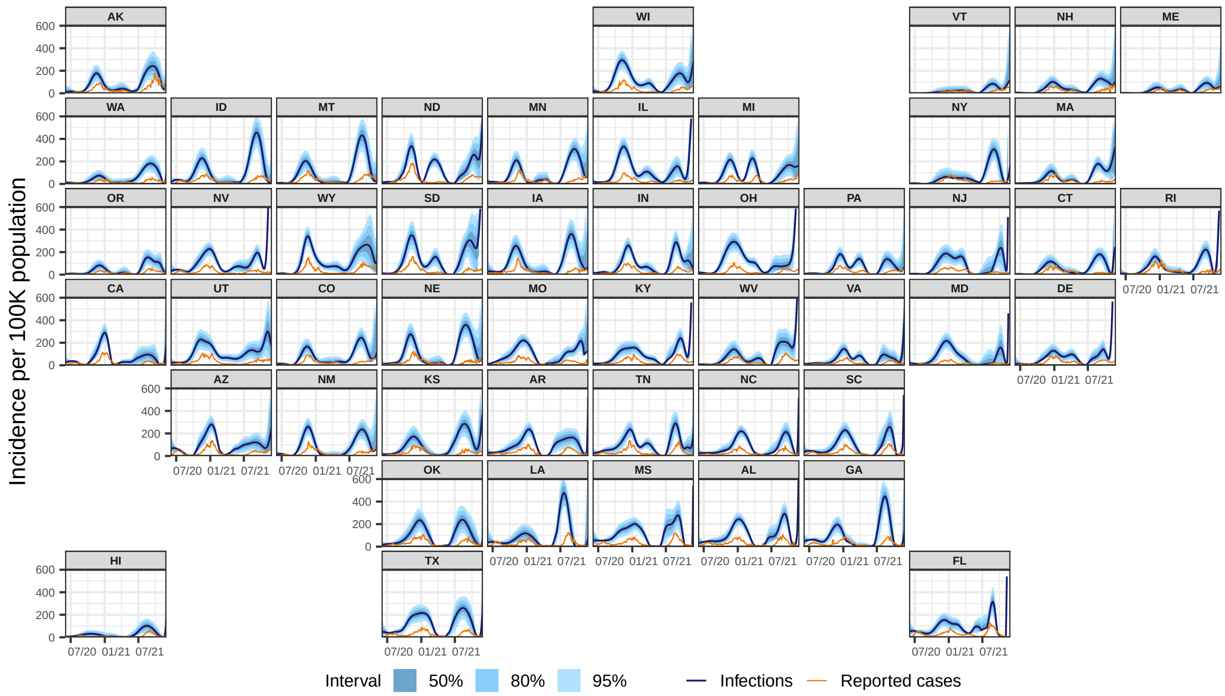

Figure 3: Estimates of the daily new infections per 100,000 population for each U.S. state from June 1, 2020 to November 29, 2021 (dark blue line). The blue shaded regions depict the 50, 80, and 95% intervals for the estimates, while the orange line represents the trailing 7-day average of reported cases per 100,000.

evident in North and South Dakota over the spring of 2021 that was virtually undetectable from reported cases. Similarly, in late-summer 2021, the Delta wave was only faintly detectable from cases in a number of Northeastern states, while infections suggested that it had already begun in earnest.

Moreover, cases severely underestimated infections during Delta for many states, more so than in earlier waves (Figure 3). The most extreme was New Jersey, where about 6.3% of estimated infections were eventually reported as cases. Similarly low were Maryland (7.3%), Nevada (8.4%), and South Dakota (10.0%). In 44 states, fewer than 1/3 of infections eventually appeared in case reports. The cases-to-infections ratio was larger in earlier waves, and its effects were most apparent in different regions. During Alpha, Louisiana had the lowest ratio of infections to cases (11.9%) followed by California (13.6%). Such patterns were less apparent during the Ancestral wave, where Ohio and Maryland had the lowest ratio of reported cases to infections at 21.4% and 21.7%, respectively.

Figure 5 shows that using cases as a proxy for infections could lead to misunderstandings in the locations that were affected and the extent to which they were affected. For example, on October 20, 2020, while case rates were elevated in a handful of upper-Midwestern states (namely, North and South Dakota), infection rates were elevated to a similar extent in the surrounding states as well, indicating a wider impact than suggested by cases alone. On July 20, 2021, while the map of case rates showed low and geographically consistent impact, infection rates revealed that Texas, Louisiana, Georgia, and their neighbors were hotspots.

By focusing on states with elevated cases, infection outbreaks may have been overlooked. For instance, on August 27, 2021, Montana and Idaho had some of the highest infection rates (Figure 5). In contrast, their case rates were unremarkable (the highest case rates tend to be in the Southeast). Infection outbreaks tended to precede case outbreaks, though the lead time could vary widely. During the Delta wave, infections in Montana peaked about 41 days before cases, while in Idaho, they peaked about 6 days before cases (Figure 3). During the Ancestral wave, infections peaked about 12 days earlier than cases in Montana and 24 days earlier in Idaho, demonstrating a notable shift in lead times.

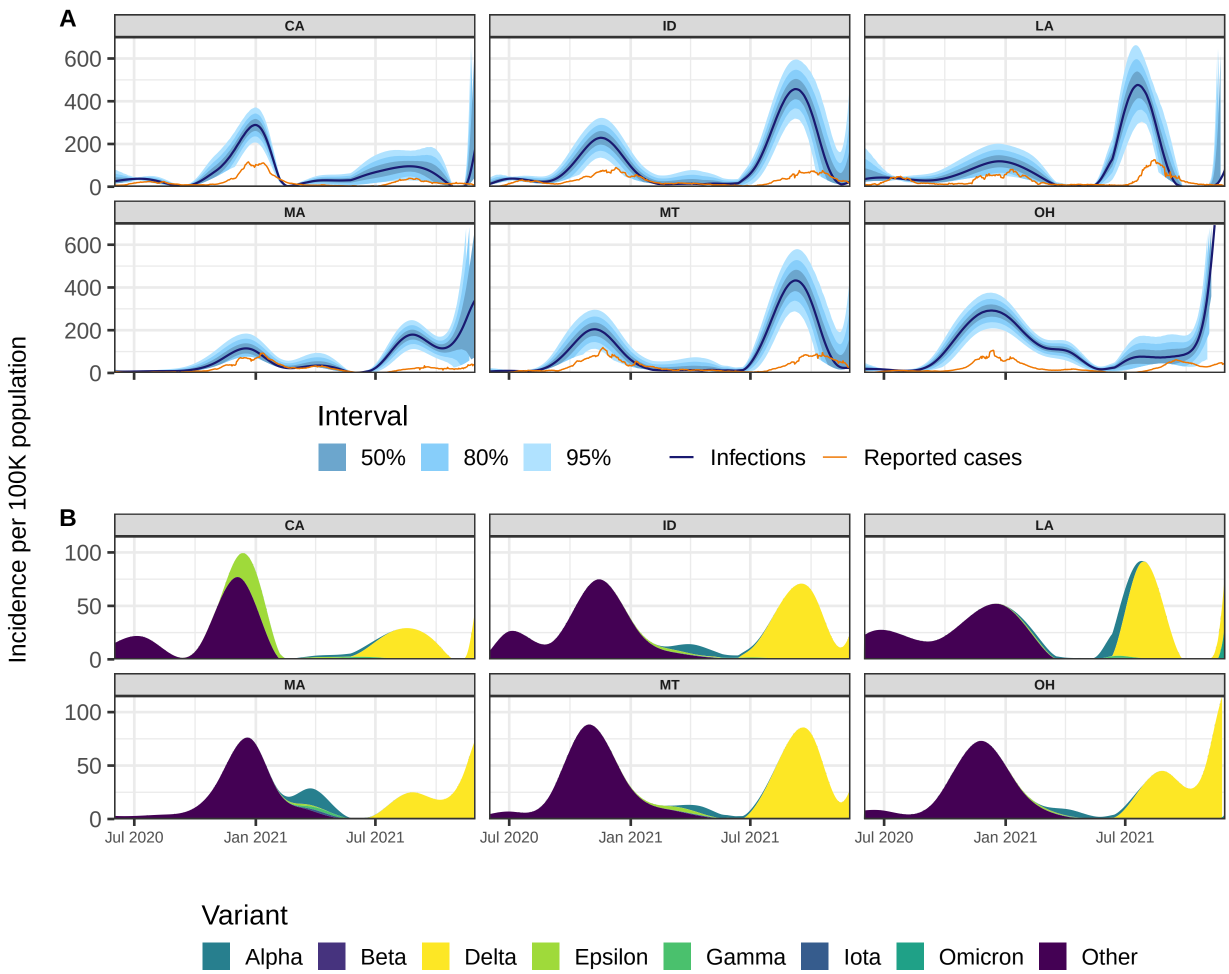

Figure 4: Panel A: Reported cases (orange) and estimates of daily new infections (dark blue) per 100K inhabitants. The blue shaded regions indicate 50, 80, and 95% confidence bands. Panel B: Deconvolved cases colored by variant per 100K inhabitants.

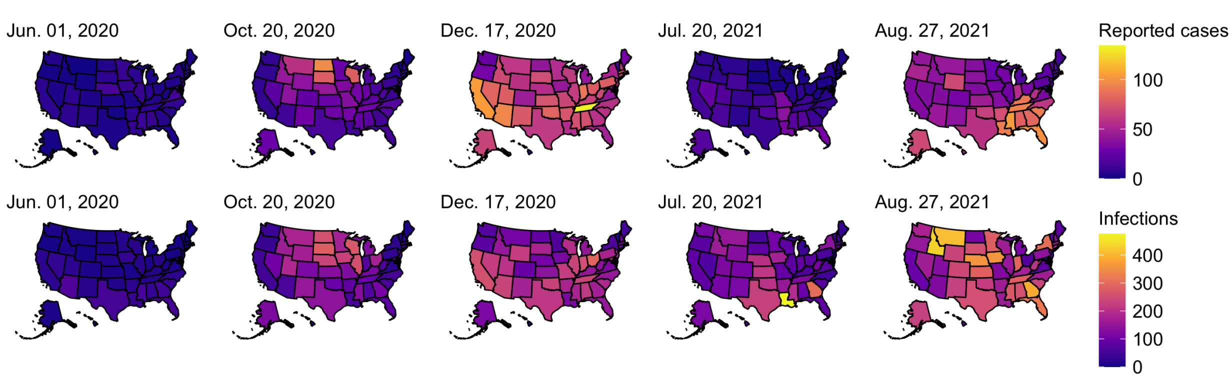

Figure 5: Choropleth maps of the state-level estimates of the daily new cases per 100K (top row) and the daily new infections per 100K (bottom row) for five select dates between June 1, 2020 and November 29, 2021. Note that the first date was chosen as a baseline, while the other dates were chosen because they present large counts of infections across all states. In particular, the third and fifth dates presented the largest number of total infections across the 50 states within those calendar years. For a more dynamic visualization of the case and infection trends over time, videos depicting the infection and case choropleth maps over the entire period of interest are available in the Github repository for this work.

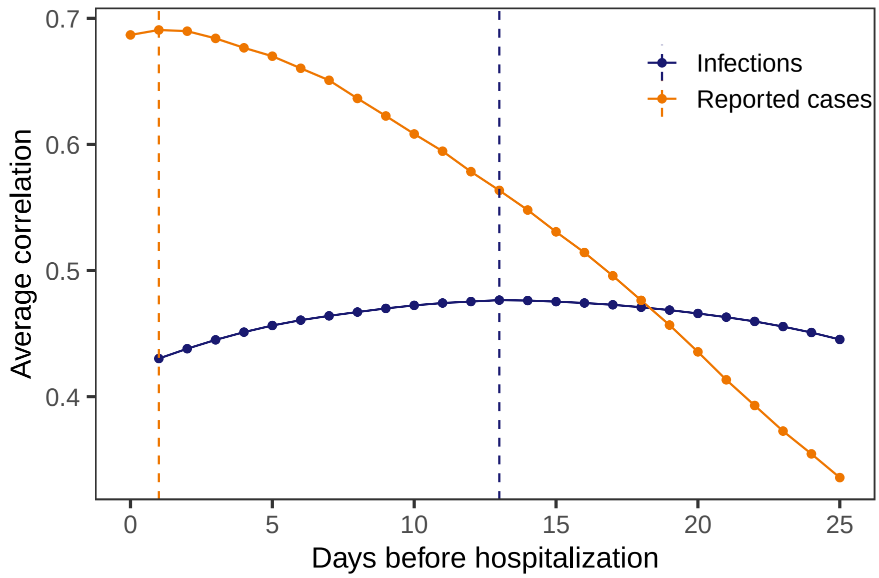

Figure 6: Spearman's rank correlation between each of infections and cases with hospitalizations per 100,000. A rolling window of 61 days was applied before averaging across all states and times for each lag. The vertical dashed lines indicated the lags for which the highest average correlation was attained.

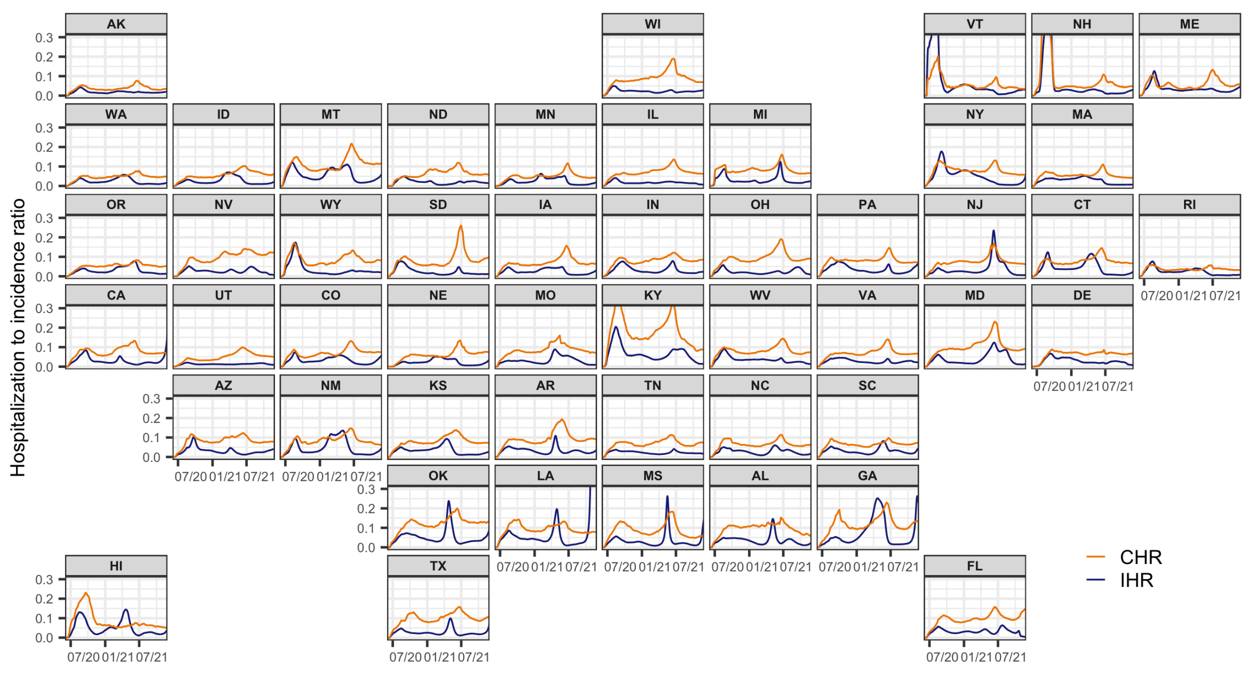

Figure 7: Time-varying IHR and CHR estimates for each state from June 2020 to November 2021, calculated using the correlation-maximizing lags from Section 3.2. Note that the infection, case, and hospitalization counts were subject to a center-aligned 7-day average to remove spurious day of the week effects. Also note that the different starting points across states were due to the availability of the hospitalization data.

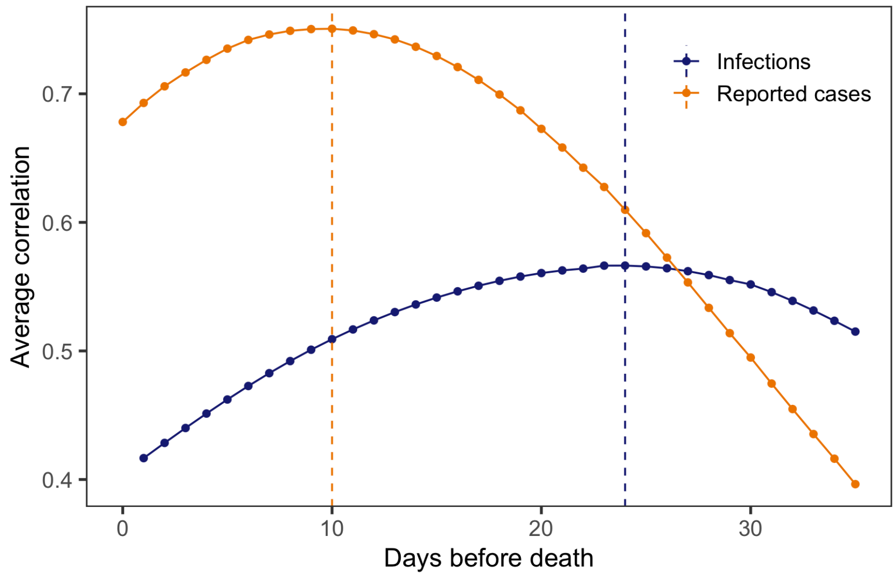

Figure 8: Spearman's rank correlation between each of infections and cases with deaths per 100,000. A rolling window of 91 days was applied before averaging across all states and times for each lag. The vertical dashed lines indicated the lags for which the highest average correlation was attained.

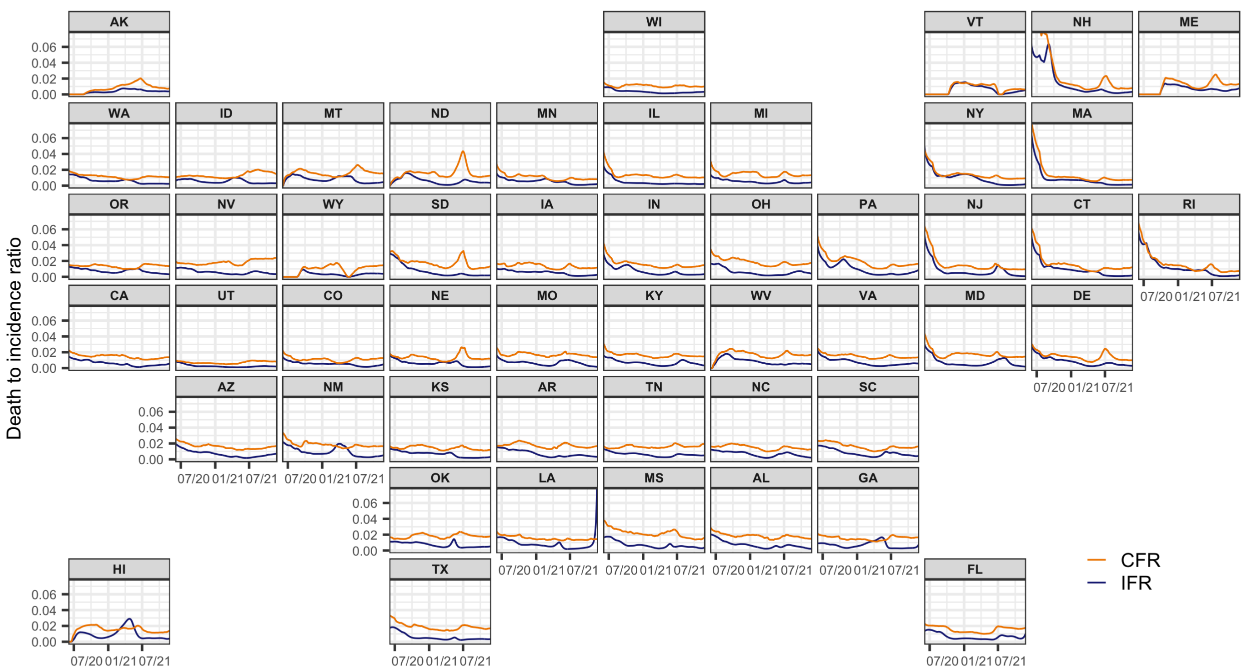

Figure 9: Time-varying IFR and CFR estimates for each state from June 2020 to November 2021, calculated using the correlation-maximizing lags from Section 3.2. Note that the infection, case, and death counts were subject to a center-aligned 7-day average to remove spurious day of the week effects.

3.2 Insights from cross-correlations, IHRs and CHRs, and IFRs and CFRs

The maximum Spearman's correlation between infections and hospitalizations was 0.48 and occurred at a lag of 13 days (Figure 6). In contrast, we found that the largest average Spearman correlation for cases was 0.69

and occurred at a lag of 1 day. That is, case reports were nearly contemporaneous to hospitalizations, while infection estimates clearly preceded them. For infections and deaths, the maximum Spearman's correlation was 0.57 and occurred at a lag of 24 days (Figure 8), whereas for cases and deaths it was 0.75 at 10 days. Thus, the maximum correlation for infections occurred 14 days earlier than for cases.

We computed the time-varying infection-hospitalization ratios (IHRs) for each state using a 13-day lag and case-hospitalization ratios (CHRs) with a 1-day lag for comparison Figure 7. Overall, the relationship between infections and hospitalizations was complex. It was characterized by intermittent spikes that punctuated longer periods where the IHRs were relatively stable, remaining below 0.1 hospitalizations per infection. This is further supported at the state-level by the mean and 95% confidence intervals for the overall IHRs described in Appendix M. The trend was also evident in the IFRs, computed using the optimal 24-day lag. These IFRs were generally more stable and smaller than the corresponding CFRs, remaining below 0.02 deaths per infection (Figure 9 and Appendix M).

While these trends tended to persist for larger window sizes, it is important to acknowledge that the window size played an important role in smoothing: larger window sizes resulted in more smoothing, leading to more tapered peaks, whereas smaller window sizes reduced smoothing, resulting in more pronounced peaks and larger intermittent spikes, as detailed in Appendices J and L.

The IHRs and CHRs exhibited similar spatiotemporal trends as those noted for infections. Namely, states that were proximate (for example, North and South Carolina) showed similar temporal patterns in IHRs and CHRs. In addition, similar spikes were evident across many states during waves of infections that were driven by variants of concern. For example, many states exhibited a striking increase in hospitalizations in mid-2021, which coincided with the rapid takeover of the Delta variant (Hodcroft, 2021). These trends were less pronounced in the IFRs and CFRs, with notable exceptions such as North and South Dakota, where there was a sharp spike in IFR and CFRs in mid-2021, also coinciding with the spread of the Delta variant.

4 Discussion

We retrospectively estimated daily incident infections for each U.S. state over the period June 1, 2020 to November 29, 2021. Our estimates support the intuition that the pandemic impacted states earlier and at a larger scale than is indicated by reported cases. They also emphasize that using cases as a proxy for infections can lead to erroneous conclusions about trends in infections. More importantly, we observed outbreaks in infections that are missed from inspecting cases alone such as the Delta wave in New Jersey, Connecticut, and Maryland. These sorts of omissions serve to emphasize that cases paint an incomplete picture of the pandemic, especially when outbreaks are largely driven by unreported infections. Furthermore, since case reports generally follow symptom and infection onsets, cases have a built-in temporal bias. This is in addition to other biases from differences in reporting across states such as temporary bottlenecks due to influxes of data or more persistent processing issues that increase the average time from case detection to report (Dunkel, 2020; Washington State Department of Health, 2020). Thus, while reported cases provide an indication of the trajectory of the pandemic, it is delayed and incomplete.

Case reporting has varied widely over time and across locations globally, not just in the U.S., leading to misrepresentations of disease burden and complicating the development of evidence-based policy decisions. This geographical variation, driven by differing testing capacities and reporting standards, has made it difficult to establish accurate infection trends and implement appropriate responses. Misleading spatial patterns of reported cases, such as overrepresentation in some areas and underreporting in others (a glimpse of which was shown in Figure 5), have further exacerbated the difficulty in evaluating the true scope of the pandemic and likely hindered effective policy decisions.

While countries like England addressed these case-based challenges through large-scale infection-prevalence studies, such as the REACT-1 study (Imperial College London, 2022), these efforts are limited by the biases inherent to the study design, specifically those related to response rates and population coverage. For example, the response rate varied between 11.7% and 30.5% in REACT-1, with demographic variation, particularly in relation to age (Elliott et al., 2023). Additionally, such studies are expensive to design and implement, requiring substantial funding for participant recruitment, data collection, and analysis. Costs are further compounded due to comprehensive national sampling as well as mass PCR-based testing (Pavelka et al., 2021). By leveraging routine data streams, our methodology provides a less expensive, more sustainable way to track infections over time. Thus, our approach has the potential to be adapted into a cost-effective, real-time alternative for pandemic monitoring, offering timely estimates to inform policymakers and overcoming the biases seen in case reporting.

Though our approach could be extended to real-time estimation, it would be complicated due to data-driven issues such as delay and right censoring. For instance, Jahja et al. (2022) explore estimating infections in real-time using the CDC line list data set in a similar deconvolution-based approach. They highlight complexities with using this data set for estimation, including monthly updates and right censoring. To handle the right censoring problem, they created a Kaplan-Meier-like method and they applied specialized regularization techniques to manage fluctuations in estimates around the date of interest. Similar techniques would likely need to be applied to manage the CDC line list dataset if we were to extend our approach to a real-time analysis. For a brief overview of the other main data sources we used, their data collection period, update frequency, and major sources of delay, which are factors that would likely impact right censoring and increase volatility in real-time estimation, please see Appendix A.

Our approach offers a number of additional advantages. By incorporating state-level case, line list, and variant circulation data, we are able to construct incubation and delay distributions that are spatiotemporally specific. Time-varying and state-specific seroprevalence data allows the reporting ratio estimates to similarly vary over space and time, a departure from existing work (Center for the Ecology of Infection Diseases, 2020; Unwin et al., 2020). Unlike previous approaches that use a single delay distribution to generate estimates for all states (Chitwood et al., 2022; Jahja et al., 2022; Miller et al., 2022), our work avoids this assumption of geographic invariance, an assumption that is far from realistic due to differences in the reporting pipelines, pandemic response, and variants in circulation, among other things. Similarly, prior methodology relies on only one incubation period distribution (Miller et al., 2022), whereas our method incorporates variant-specific incubation periods. This enhances our infection onset estimation by accounting for the differences across variants–specifically, that newer variants tend to have shorter incubation periods (Ogata et al., 2022; Tanaka et al., 2022; Wu et al., 2022).

Another limitation of previous approaches to estimate infections is that they often fail to account for reinfections. While reinfections constitute a small portion of the total infections until the arrival of high immune-escape variants (Omicron BA.1), disregarding them means that the infection-reporting ratio will tend to be underestimated with seroprevalence data alone. By accounting for reinfections as well as the waning of seropositivity, we more accurately estimate this ratio. However, future work could refine this analysis. Because the waning of immunity is likely to be variant-dependent (Pooley et al., 2023), our model's single waning parameter would be more accurately estimated as a mixture of variant-specific parameters with weights determined by the proportion of the variants circulating.

While we did consider the waning of seropositivity in our modelling procedure, we did not explicitly consider the relationship between the epidemic growth rate and the probability of testing positive in PCR tests. Specifically, during high rates of growth in an epidemic, a greater proportion of infections come from individuals early in their infection, who have higher viral loads and are more likely to test positive. During a decline, more individuals are later in their infection, having lower viral loads, which makes them less likely to test positive. This dynamic relationship between infection stage, viral load, and test positivity has been well-described by Hay et al. (2021). In the context of the data sources that we used, since viral RNA loads tended to peak early on infection (Frediani et al., 2024), individuals in the early stages of infection were more likely to test positive and, therefore, be represented in case reports. We acknowledge that this effect may be substantial and could influence the accuracy of the deconvolve case and infection estimates, especially during rapid changes in the epidemic trajectory. Thus, future work could explore how incorporating how viral loads change over time could the improve the precision of our infection estimates, as incorporating individual-level viral load data could help to refine the timing of infections relative to case reports.

In addition to incorporating viral load data, another area for potential improvement is the lag time between infections and hospitalizations. While REACT-1 has suggested that the lag between infections and hospitalizations changes over time (Eales et al., 2022), we used a single lag because our initial goal was to find a lag that maximized the correlation between infection and hospitalization rates across all states over a considerable period of time. Moreover, using a single lag simplified the estimation process, relying on few modelling assumptions and helping to stabilize the ratios (especially in comparison to potentially overfitting by varying the lags for each state). Nevertheless, future work could explore time-varying lags, particularly with respect to how well they align with the emergence of new variants and policy measures.

Another aspect of our analysis that warrants consideration is the time period under study. We chose to end our analysis on November 29, 2021, for two main reasons. The first is that Omicron and subsequent variants come with substantial increases in the risk of reinfection in comparison to previous variants, likely due to increased immune escape (Eythorsson et al., 2022; Pulliam et al., 2022; Wei et al., 2024). Access to reinfection data that is representative of each location under study is paramount for extending the analysis. While it would be ideal to use the reinfection rates over time for each U.S. state, many states do not publicly report reinfection data over the entire time period under examination, if at all.

The second reason is that the case-ascertainment ratio after December 2021 can no longer be estimated with seroprevalence data alone. Specifically, while most state-level data suggests that reinfections still account for less than 20% of reported cases during Omicron (Hawaii Department of Health, 2022; New York State Department of Health, 2023; Ruff et al., 2022; Washington State Department of Health, 2022), seropositivity rapidly reaches nearly 100% of the population. Therefore, alternative data sources for estimating the case-ascertainment ratio must be considered. For example, wastewater surveillance data may be complementary to seroprevalence data, especially when testing is low, or serve as a substitute when it is unavailable (McManus et al., 2023). An alternative approach could integrate surveillance streams from surveys, helplines, or medical records if they offer a sufficiently strong signal of the disease intensity over time (European Centre for Disease Prevention and Control, 2020; Reinhart et al., 2021).

Our work develops a deconvolution-based approach to inferring infection onset, combining available line list data with variant circulation estimates and literature derived incubation periods. This approach is complemented with the development of a model that incorporates waning detectable antibody levels and major seroprevalence surveys. The resulting infection estimates as well as their geospatial and temporal trends are strongly grounded in both data and statistical models.

These well-informed, localized estimates of COVID-19 infections provide a clear and comprehensive understanding of the pandemic's progression over time. They contribute important information on the timing and magnitude of the disease burden for each location, and highlight trends that may not be visible from reported case data alone. Therefore, these infection estimates provide key information for the ongoing investigation on the true size and impact of the pandemic.

CRediT authorship contribution statement

Ryan J. Tibshirani: Conceptualization, Methodology, Writing - Reviewing and editing. **Daniel J. McDonald:** Conceptualization, Methodology, Software, Visualization, Writing - Reviewing and editing. **Ajitesh Srivastava:** Methodology, Writing - Reviewing and editing. **Rachel Lobay:** Methodology, Software, Visualization, Formal analysis, Writing - Original draft, Writing - Reviewing and editing.

Data availability

The required materials and code for reproducing the figures and the numerical results are available at https://github.com/cmu-delphi/latent-infections/.

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Competing interests

The authors declare no competing interests.

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